

ConventionCorp: Summary Paper

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In conversations with RIT professors and TAs, a lack of engaging methods of coding convention education was identified. It was observed that students often fail to follow good coding practices as their RIT careers go on, especially if they are not incentivized by grade points. Furthermore, a lack of interest and engagement in learning these practices when they are first taught was also reported. Thus, we set out to create a tool to engage students in coding convention education as well as reinforce the concepts of what these practices are and why they are important.

Gamification was quickly identified as the best tool to create this engagement. There has been extensive research into the topic of games being used as educational tools. According to a review of such literature by Zeybek et al. [2023], most of this research has found that gamification is a potent tool to increase motivation to engage with a topic, especially one where there is little motivation before the game is introduced. This research aligned with our own observations and thus we created ConventionCorp, a gamified tool for coding convention education.

In the game, the learner takes the role as a project lead at the titular company. The company, whose role is to put on intergalactic conventions, has just hired a new employee that comes from a far-off planet where they don't use variable names; they just refer to memory addresses directly. To acclimate this employee to the company, you must teach him all that you have learned about naming conventions. This is done so in a choose your own adventure style dialogue tree. You can give him good advice or bad advice, and he will develop his code accordingly. This means that should you give him bad advice, whether intentionally or not, he will come back to you with barely readable code that follows the path you have led him down.

This form of negative feedback was an intentional decision that we theorize will lead to a positive impact. By giving students a direct and comical example of what happens when these rules aren't followed, they will not only reinforce their understanding of naming conventions, but also why they exist. Furthermore, we hope that the comedic aspect of both the dialogue tree and the over-the-top variable names will lead to increased engagement with the game as there is an added entertainment value on top of the educational value. Another intentional decision was to put the student in the position of teacher. It is well known that putting a learner in the shoes of a tutor greatly reinforces their own understanding of the material. We believe that our game takes advantage of this long proven educational phenomenon.

The implementation of the project was done in Godot. The main gameplay loop is controlled by a set of array-based trees. There are two trees, one for the dialogue displayed in the

textbox at the bottom of the screen, and one for the code displayed on the computer once clicked. At each node in the computer tree, a variable containing the text that should be displayed at that point is contained. The same is true for the dialogue tree, except there can also be a subtree with additional dialogue that doesn't affect the main gameplay loop. The main gameplay loop is controlled by a universal controller that represents where the overall gameplay tree the player is. The program outputs whatever variable is at that node in the respective trees to the respective textboxes. All input is through Godot's built-in key press system or button click system.

There are still further directions this project can go. For example, the main screen was AI generated and is only one still image. Human-generated visuals and animations could create even more engagement in the game. Furthermore, while the game was made with researched based pedagogy in mind, it has yet to be user tested on students who are first learning these concepts. Lastly our demo focuses on teaching naming variable conventions, though the program was built to be adaptable to other types of content such as good commenting practices and could be adapted for those purposes by only changing the content and not having to even read through the full codebase.

References:

[1] Nilufer Zeybek and Elif Saygi. 2023. Gamification in education: Why, Where, When, and How? - A Systematic Review.

<https://journals.sagepub.com/doi/full/10.1177/15554120231158625>

GitHub Link:

<https://github.com/borebagger42/ConventionCorp>

Poster:

ConventionCorp: A Novel Approach to Coding Convention Education

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Motivation

In conversations with RIT professors, TAs, and instructors within the Software Engineering department, an obvious need for improved teaching methods surrounding coding conventions was identified. It was clear that students were not demonstrating good coding practices, especially in courses where they weren't explicitly taught. The cause of this lack seemed to stem from one of two issues: A lack of motivation to understand why these are necessary or an understanding of how to implement them. Thus, we set out to create an educational tool that addresses both of these issues.

What is ConventionCorp?

ConventionCorp is a game-based educational tool for teaching coding conventions. The user plays as an employee at the titular company, which runs a logistics website for intergalactic conventions. They are responsible for onboarding a new employee who comes from a planet where our naming conventions don't exist. It is structured and styled like an old-fashioned pixelated choose-your-own-adventure game, where you can give the new employee good advice or bad advice, and the dialogue tree will progress accordingly.

Why ConventionCorp?

There has been a large amount of research into the benefits of gamification of instructional material. Across several studies, it has been shown to increase motivation for students to engage with the content. Given that we identified motivation as part of the cause, gamification was a clear solution. In addition, we implemented several other game features meant to achieve our goals of increasing understanding and motivation. The following three stand out as most impactful:

1. **Freeform Dialogue Tree:** You can give the employee good or bad advice. You are not immediately told that your response is incorrect, but the consequences of bad advice will be felt. The reasons for conventions are shown rather than told.
2. **Tone & Aesthetic:** The lighthearted tone and pixelated art style are meant to evoke nostalgic feelings of classic video games. This keeps the student engaged beyond its educational value.
3. **Learner as Expert:** Our learner is put in the shoes of the expert, which has been shown to increase understanding of the material.

Example



Implementation

This project was created in Godot, a lightweight game engine. There is a dialogue tree for both the literal dialogue and the code that appears on the computer screen. These are both stored in an array. Through button presses, the user manipulates the universal controller, which determines what data from the trees needs to be read and displayed into the computer and dialogue.

Future Work

Our example was built to teach variable naming conventions; modularity was a primary driver for implementation details. The universal controller handles all progression, so in order to adapt this to a different subject material, all you have to do is change the contents of the dialogue trees. Thus, other modules, such as commenting practices, would require no knowledge of the program to create.

Furthermore, there could be improvements made to the interactivity, especially in regards to the humanoid character and additional reward mechanisms for doing a good job, such as office customization.